

1 Summary

Model (Run-name): model1 (test)

Iteration: 7

TAC: 152029.41590612743

Took: 1.2883949279785156s

Total Time: 5.260713577270508s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

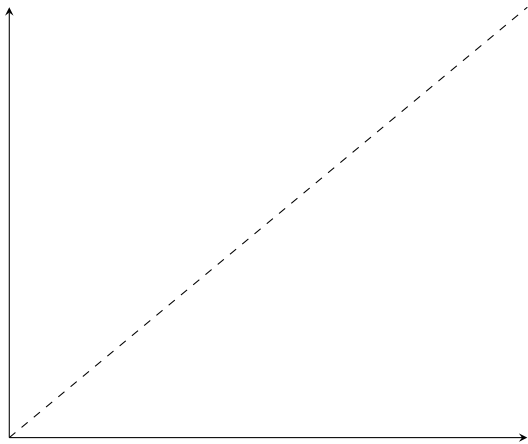
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

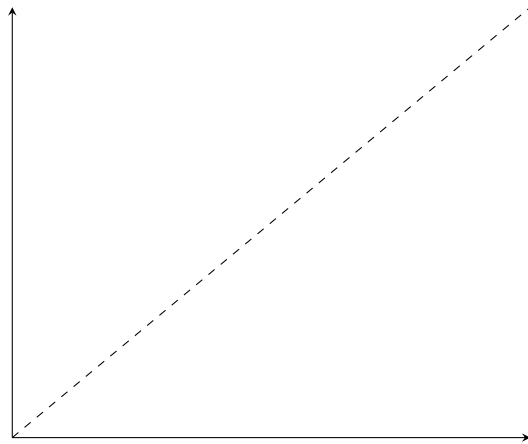
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05347468613135327	$b_{1,1,1}^{H,out}$	846.9993906566301
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.01641493763832621	$b_{1,2,2}^{H,out}$	3909.027488151723
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.03132852964088974	$b_{2,1,2}^{H,out}$	5440.834316558422
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014179816115632665	$b_{1,1,1}^{C,in}$	8638.541232227606
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009686644743776533	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022813475154884494	$b_{2,1,2}^{C,in}$	4251.257938388832
$A_{1,1,1}^\beta$	65.40595072874316	$q_{1,1,1}$	640.9725118482606	$b_{1,1,1}^{C,out}$	9279.513744075872
$A_{1,2,2}^\beta$	64.01825678947372	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	147.48374761404892	$q_{2,1,2}$	2488.541232227609	$b_{2,1,2}^{C,out}$	6739.79917061644
$A_{cu,1}^\beta$	4.432280503220532	$q_{cu,1}$	209.02748815173516		
$A_{cu,2}^\beta$	36.670233878040335	$q_{cu,2}$	1911.4587677723912		
$A_{hu,1}^\beta$	12.544341108707222	$q_{hu,1}$	470.48625592413055		
$A_{1,1,1}$	65.40595072874316	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	64.01825678947372	$t_{1,2}^{(H)}$	585.9027488151736		
$A_{2,1,2}$	147.48374761404892	$t_{1,3}^{(H)}$	390.90274881517354		
$A_{cu,1}$	4.432280503220532	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	36.670233878040335	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	12.544341108707222	$t_{2,3}^{(H)}$	465.57293838861955		
$dt_{1,1,1}$	31.365750394942037	$t_{1,1}^{(C)}$	618.634249605058		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	575.9027488151743		
$dt_{1,2,2}$	85.90274881517394	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	40.902748815173936	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	14.097251184826096	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	55.57293838861956	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	70.90274881517351	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	145.57293838861955	$t_{1,2,2}^H$	390.90274881517337		
$dt_{hu,1}$	61.36575039494204	$t_{2,1,2}^H$	404.8354409523366		
$f_{1,1,1}^H$	2.2891875423152164	$t_{1,1,1}^C$	618.6342496050581		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	13.439619574213637	$t_{2,1,2}^C$	642.3445711002439		
$f_{1,1,1}^C$	15.0	$b_{1,1,1}^{H,in}$	1487.9719025048907		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5859.027488151699		
$f_{2,1,2}^C$	10.368921800948371	$b_{2,1,2}^{H,in}$	7929.3755487860335		

3 *RecLMTD* Tangent Points

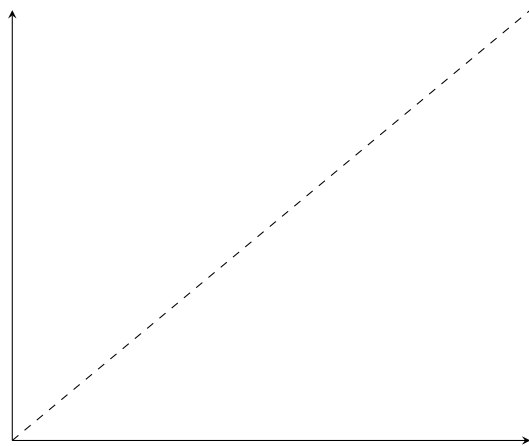
- $(1, 1, 1)$
 - Value: 0.053475
 - Error: 0.000028300271349
 - Relative error: 0.000528947508380
- $(1, 2, 2)$
 - Value: 0.016415
 - Error: 0.000074363687663
 - Relative error: 0.004509814345243
- $(2, 1, 2)$
 - Value: 0.031329
 - Error: 0.001744256578494
 - Relative error: 0.052739934486434
- $cu, 1$
 - Value: 0.014180
 - Error: 0.000014565349664
 - Relative error: 0.001026134861831
- $cu, 2$
 - Value: 0.009687
 - Error: 0.000001770571743
 - Relative error: 0.000182751428892
- $hu, 1$
 - Value: 0.022813
 - Error: 0.000002956069874
 - Relative error: 0.000129558818574



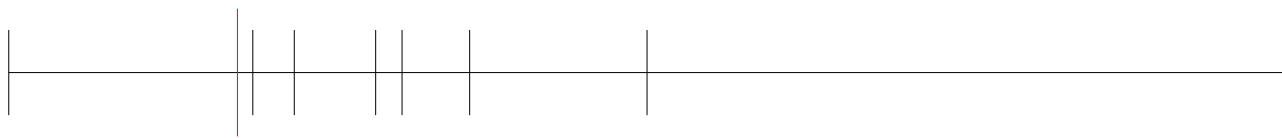
(a) *RecLMTD* - (1, 1, 1)



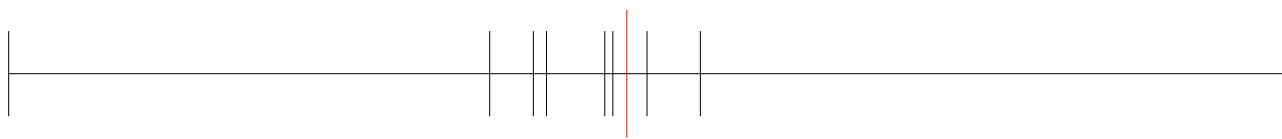
(b) *RecLMTD* - (1, 2, 2)



(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1487.9719025048907
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5859.027488151699
 - Absolute error: 3.728928277269006e-11
 - Relative error: -6.364415058317669e-15
- (2, 1, 2)
 - Value: 7929.3755487860335
 - Absolute error: 1.2732925824820995e-11
 - Relative error: -1.6057917482254139e-15

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 846.9993906566301
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3909.027488151723
 - Absolute error: 1.0459189070388675e-11
 - Relative error: -2.675649916018879e-15
- (2, 1, 2)
 - Value: 5440.834316558422
 - Absolute error: 1.000444171950221e-11
 - Relative error: -1.8387697800418342e-15

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8638.541232227606
 - Absolute error: 7.275957614183426e-12
 - Relative error: -8.422669312544545e-16
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4251.257938388832
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9279.513744075872
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6739.79917061644
 - Absolute error: 79.37854361428981
 - Relative error: 0.011917947538099861



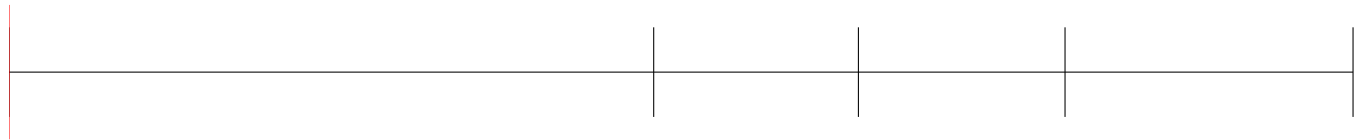
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



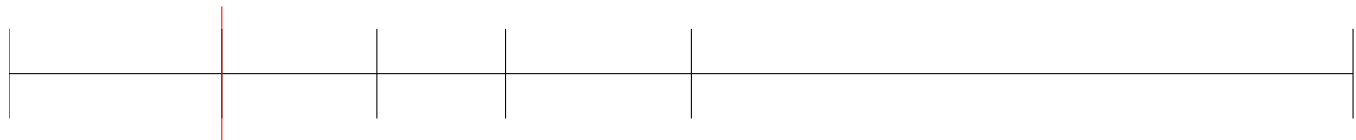
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



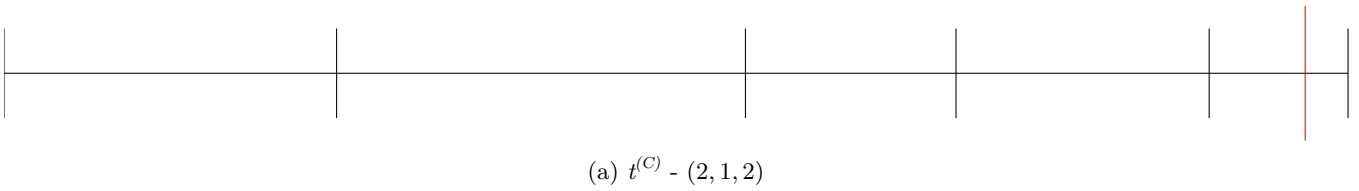
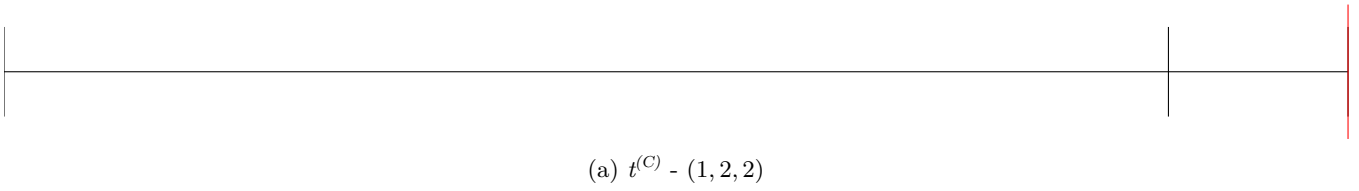
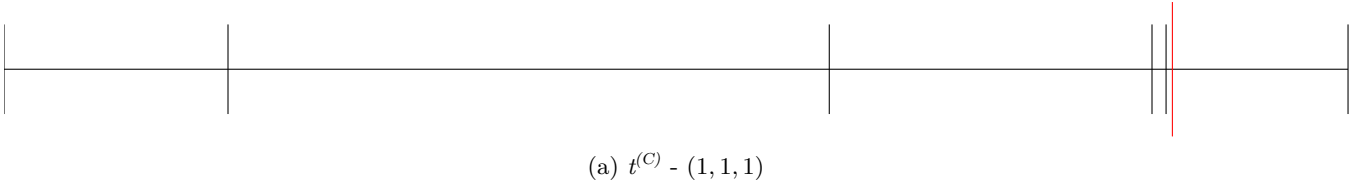
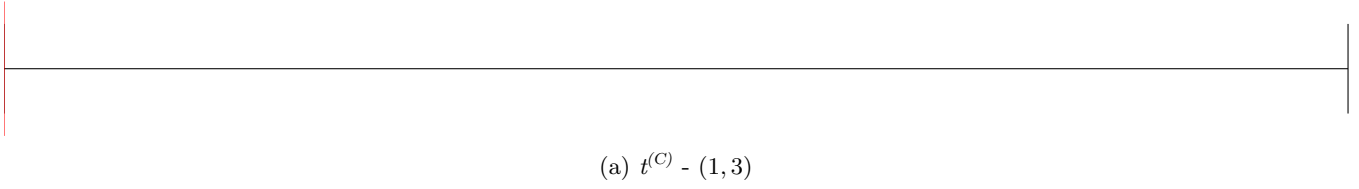
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

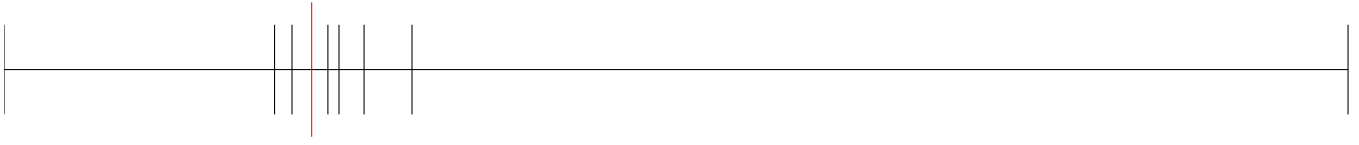
- $(1, 1, 1)$
 - Value: 65.405951
 - Error: 3.145657051078544
 - Relative error: 0.045887429236991
- $(1, 2, 2)$
 - Value: 64.018257
 - Error: 0.000000000001506
 - Relative error: -0.000000000000024
- $(2, 1, 2)$
 - Value: 147.483748
 - Error: 8.440927898788914
 - Relative error: 0.054134651048826

5.2 A_{CU}

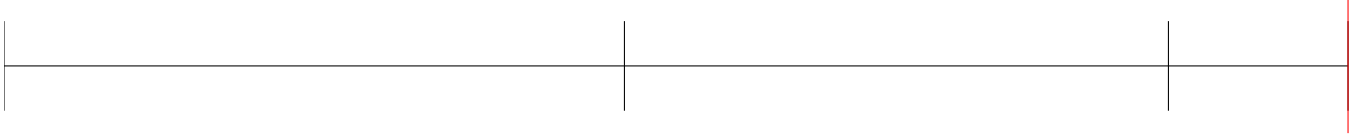
- $cu, 1$
 - Value: 4.432281
 - Error: 1.495662186987848
 - Relative error: 0.252307126628998
- $cu, 2$
 - Value: 36.670234
 - Error: 0.361010173535668
 - Relative error: 0.009748799501115

5.3 A_{HU}

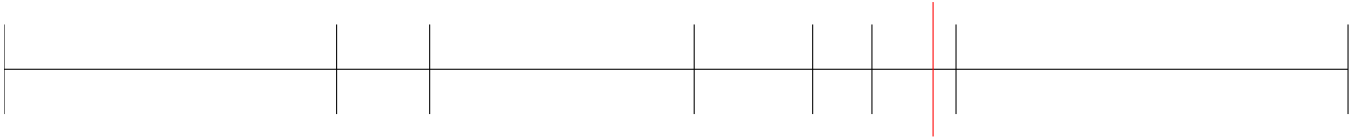
- $hu, 1$
 - Value: 12.544341
 - Error: 0.335770703580513
 - Relative error: 0.026068927698297



(a) $A^\beta - (1, 1, 1)$



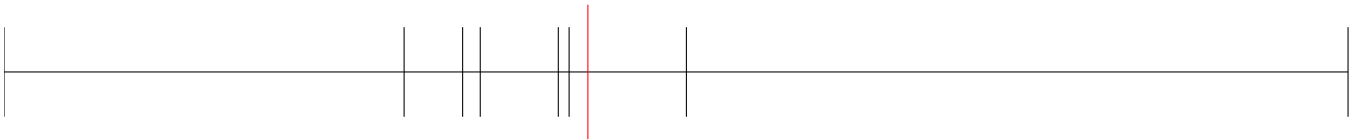
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

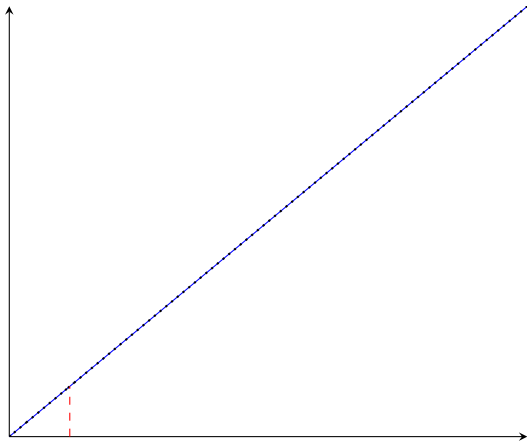


(a) $A^\beta - hu, 1$

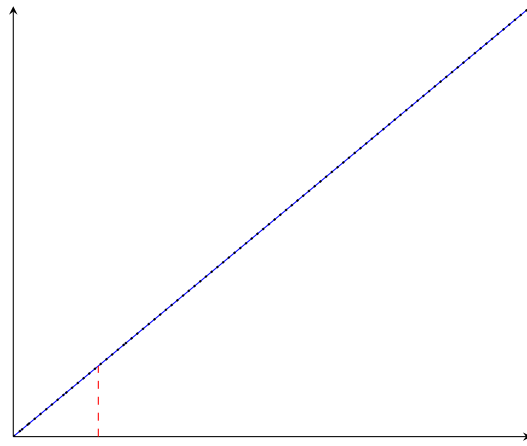
6 A^β Breakpoints

- (1, 1, 1)
 - q : 65.405951
 - $\hat{q}^\beta$: 65.405951
 - q^β : 65.405951
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, **63.3468696286352**, **69.14107665413653**, 69.89902342903967, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 64.018257
 - $\hat{q}^\beta$: 64.018257
 - q^β : 64.018257
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, **39.36524453694061**, **84.1468297226002**, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 147.483748
 - $\hat{q}^\beta$: 147.483748
 - q^β : 147.483748
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, **87.16107999267206**, **720.0**]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.432281
 - $\hat{q}^\beta$: 4.432281
 - q^β : 4.432281
 - Breakpoints: [0, **2.9819181007907165**, **7.702824927089129**, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 36.670234
 - $\hat{q}^\beta$: 36.670234
 - q^β : 36.670234
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, **34.23465128608717**, **38.88483572219945**, 285.4001551947793]

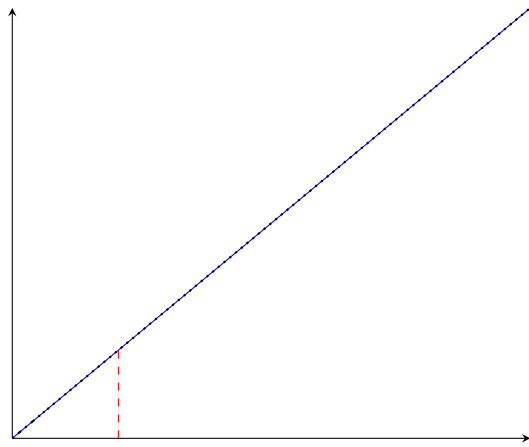
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 12.544341
 - $\hat{q}^\beta$: 12.544341
 - q^β : 12.544341
 - Breakpoints: [0, 4.943755299006499, 12.595096067947068, 13.192252946941991, 13.244985069124171, 13.30871858348944, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



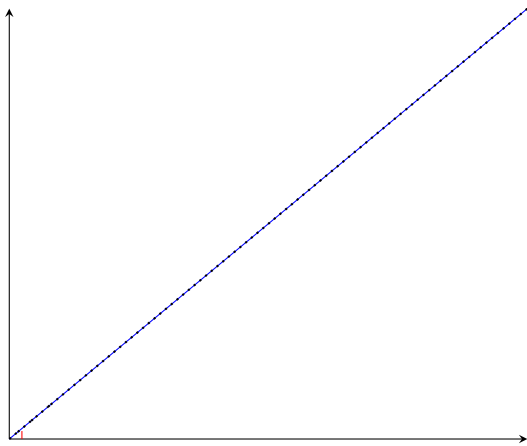
(a) $q^\beta - (1, 1, 1)$



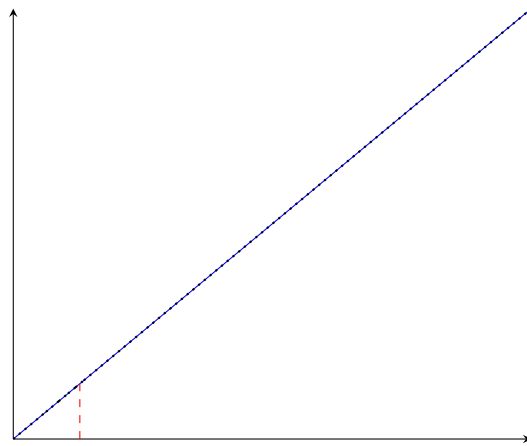
(b) $q^\beta - (1, 2, 2)$



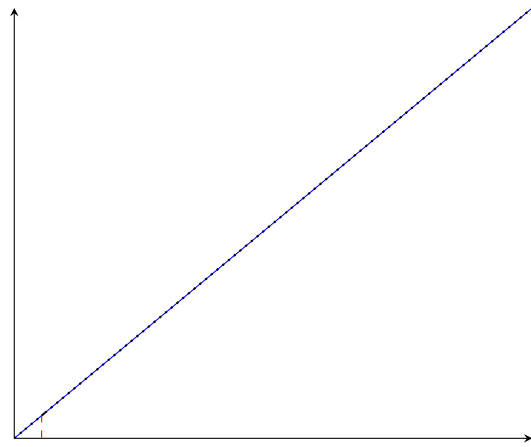
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	13.06	$b_{2,2,2}^{H,out}$	3870.624451213972
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.06	$b_{1,1,2}^{C,in}$	1898.7420616111679
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.16	$b_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	0.0	$RecLMTD_{1,2,1}$	0.016070473816068356	$b_{2,1,1}^{C,in}$	0.0
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.16	$b_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.020111728838038186	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.16	$b_{1,1,2}^{C,out}$	1898.7420616111694
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.06	$b_{2,1,1}^{C,out}$	0.0
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.06	$b_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.06	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.0		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	590.0		
$dt_{1,2,1}$	280.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	410.0		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	5012.028097495109		
$f_{1,2,1}^H$	7.710812457684783	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	3870.6244512139656		
$f_{2,2,2}^H$	6.560380425786363	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.631078199051629	$b_{1,2,1}^{H,out}$	5012.028097495106		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0		